Supplemental Information: Kin Group Gini: An Extended Family Approach to Income and Wealth Inequality in the U.S.

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Appendix A: Comparison with External Estimates

Appendix A1. Comparison of income Gini coefficients

Year	FRED	U.S. Census	Our Results (All HH)
1969	0.391	0.349	0.401
1979	0.404	0.365	0.398
1989	0.431	0.401	0.439
1999	0.458	0.429	0.465
2009	0.468	0.443	0.464
2019	0.484	0.454	0.470

Appendix A2. Comparison of wealth Gini coefficients with external estimates

Year	SCF	PSID	Our Results
1983	0.760		
1984		0.739	0.755
1989	0.781	0.742	0.751
1992	0.770		
1994		0.732	0.749
2003		0.810	0.786
2004	0.809		
2007		0.830	0.801
2009		0.890	0.833
2010	0.846		

Appendix A2. Comparison of wealth Gini coefficients with external estimates

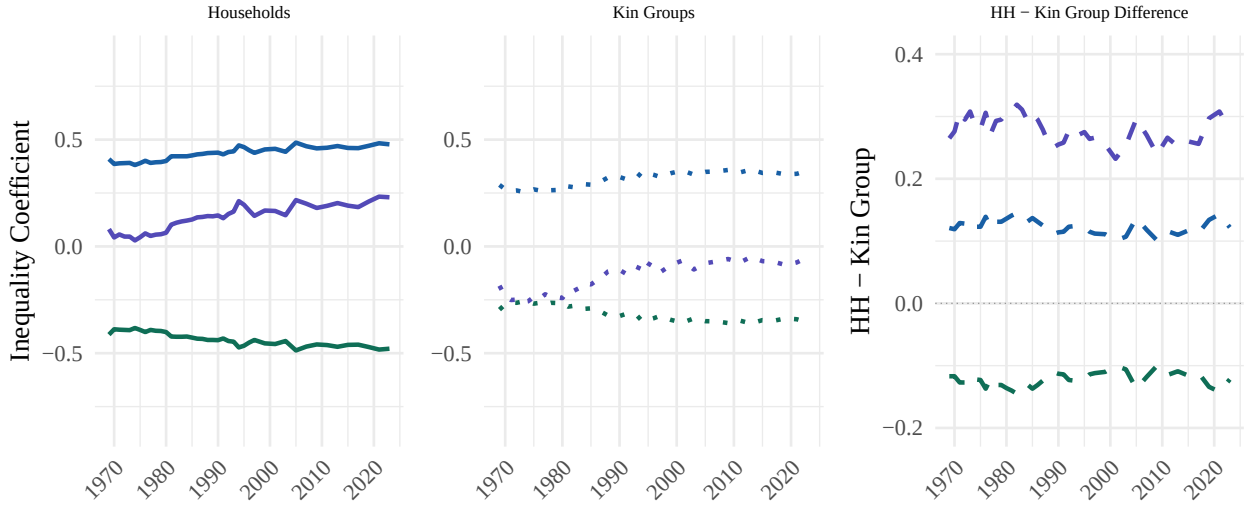
Year	SCF	PSID	Our Results
2011		0.880	0.811

Appendix A compares Gini coefficients at the household level calculated for income and wealth with Gini coefficients from other researchers. Gini coefficients in Appendix A are calculated using the full sample of households in the PSID, rather than just the households who belong to a kin group with at least one other household (which are the results presented in the main paper).

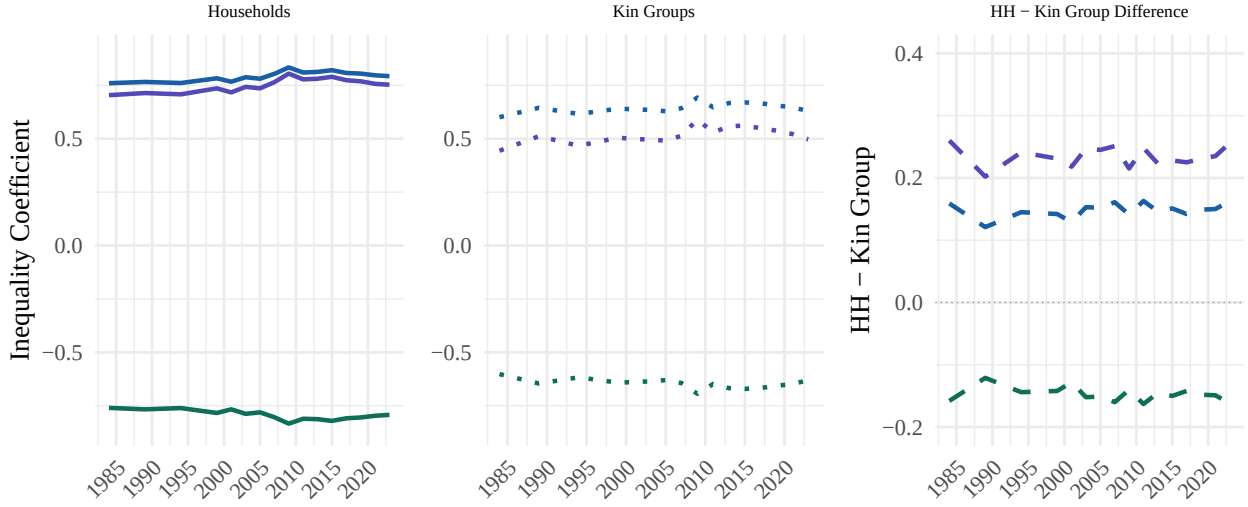
Gini coefficients are not available for all years in other sources, so only years available are presented. Gini benchmarks for income rely on estimates from the Federal Reserve Bank of St. Louis (FRED) and the U.S. Census [1, 2]. FRED uses data from the World Bank on U.S. households and the Census uses data from the Current Population Survey (CPS). Gini benchmarks for wealth rely on estimates from research papers based on data from the PSID or Survey of Consumer Finances (SCF) [3, 4].

Appendix B: Alternative Inequality Measures

Panel A: Income



Panel B: Wealth (excl. home equity)

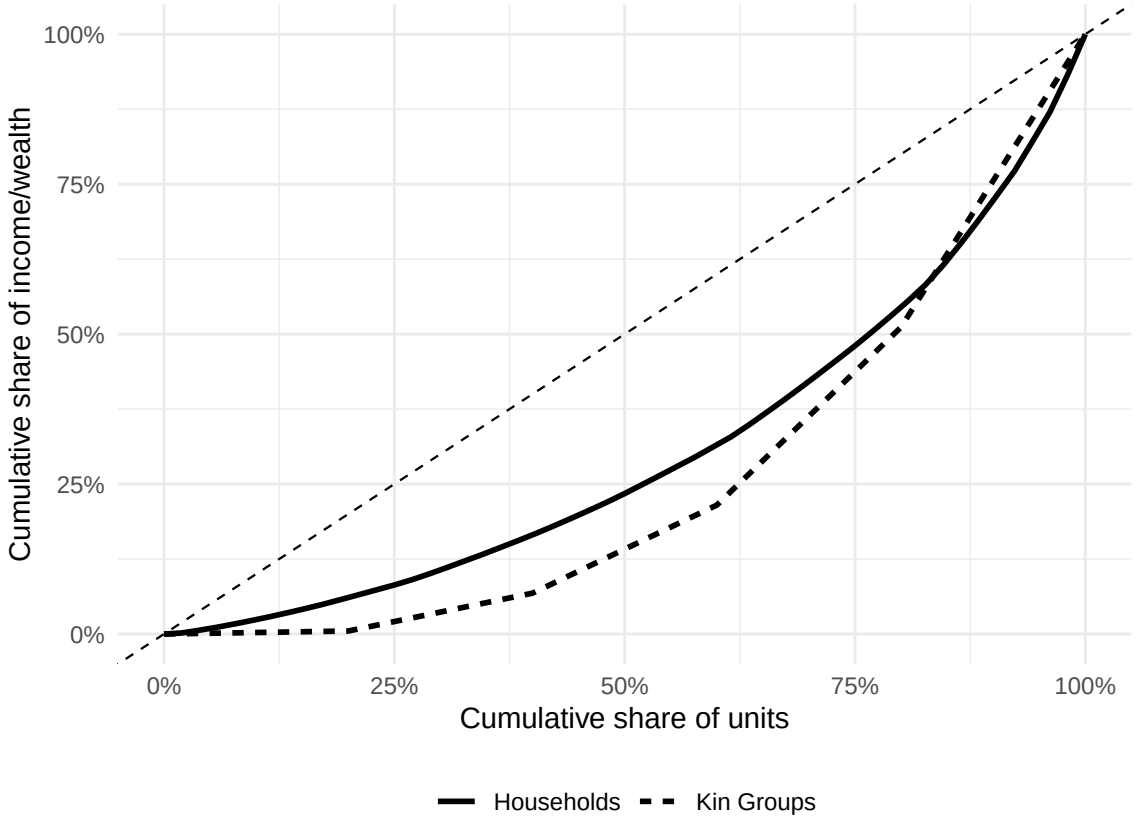


C1 (Bonferroni) C2 (Gini) C3 (Upper Tail)

Appendix B shows three alternative measures of inequality proposed by Aaberge [5]. C1 is the Bonferroni coefficient (greater weight on lower tail); C2 is the standard Gini coefficient; C3 places greater weight on the upper tail of the distribution. All estimates are weighted averages across all available years. Inequality is lower at the kin group level using any measure of inequality.

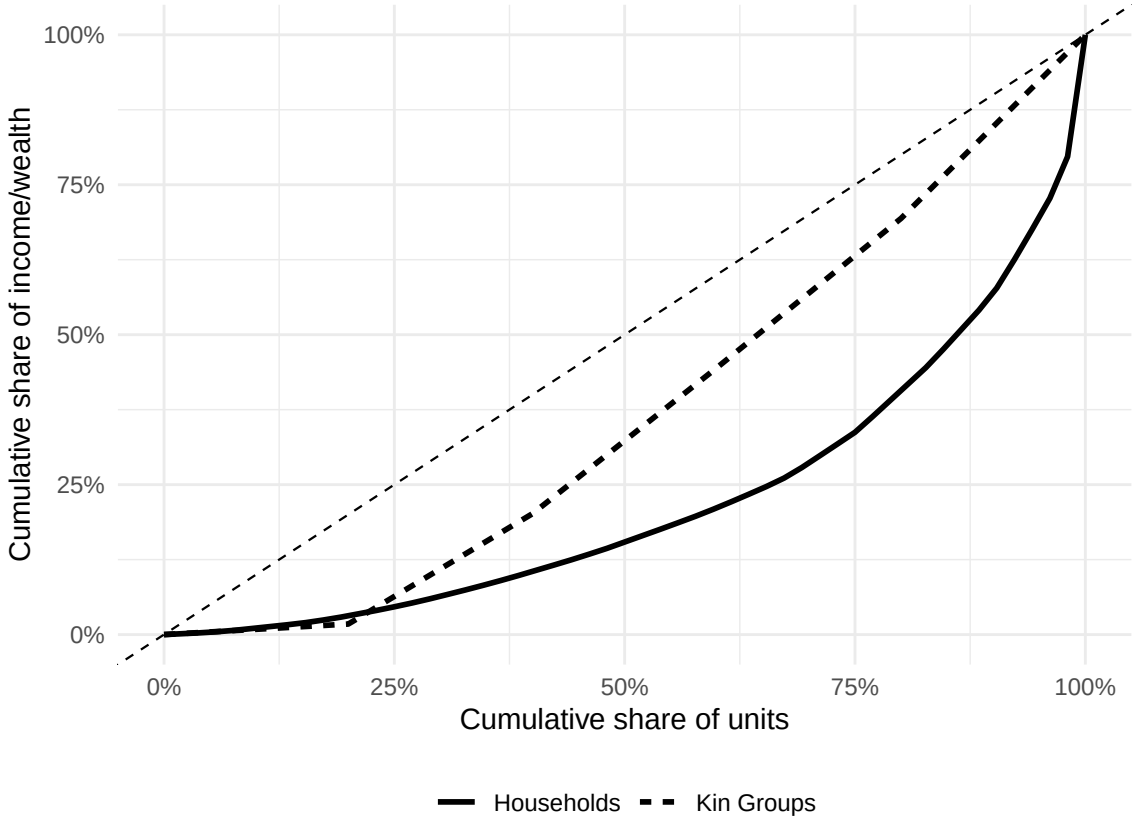
Appendix C: Simulating Gini Coefficients

Appendix C1: Case with lowest simulated kin group-household Gini difference



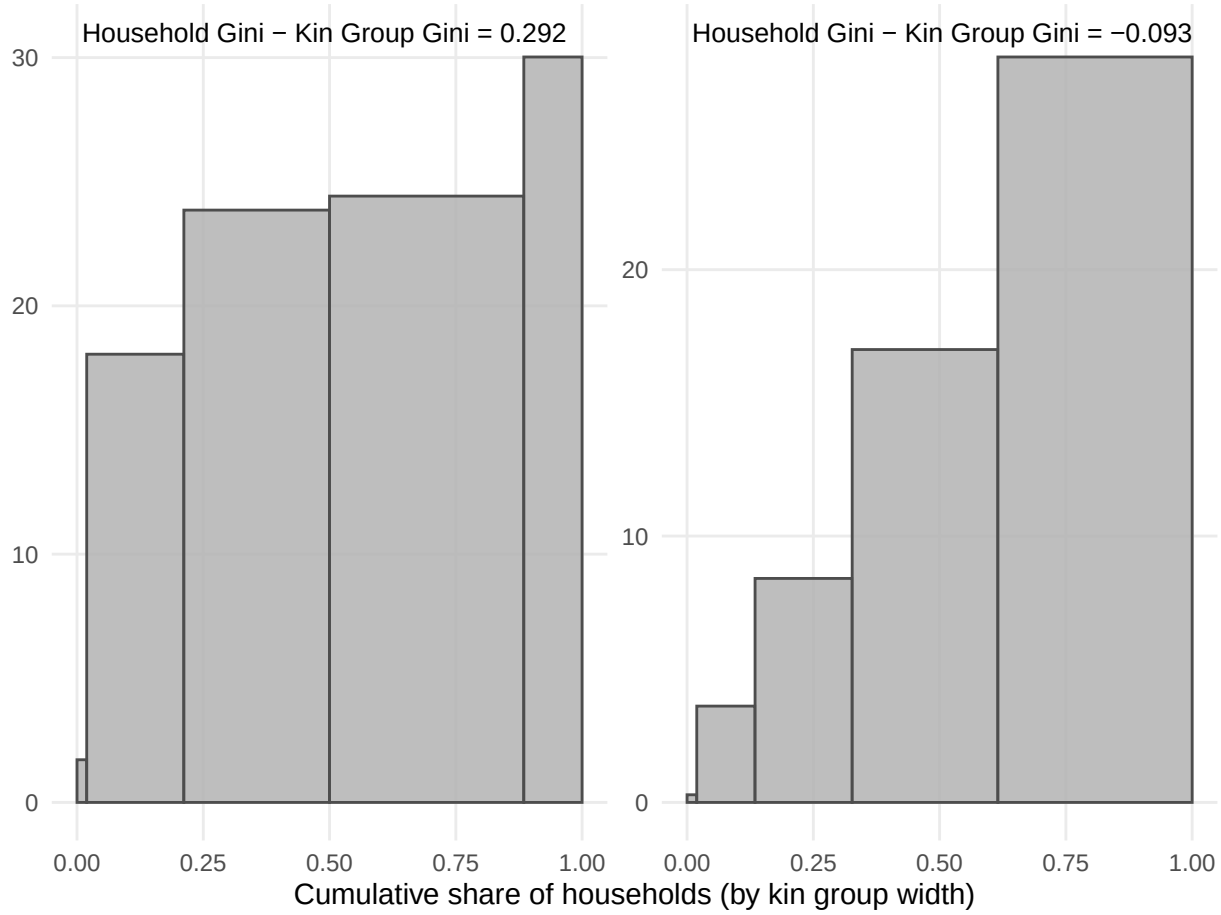
Appendix C1 illustrates the minimum possible difference between kin group and household Gini coefficients in the simulated data. This case demonstrates that Gini coefficients at the kin group level can exceed those of households.

Appendix C2: Case with highest simulated kin group-household Gini difference



Appendix C2 shows the maximum possible difference between kin group and household Gini coefficients in the simulated data. This case also demonstrates that Gini coefficients at the kin group level can exceed those of households.

Appendix C3: Summary of extreme cases



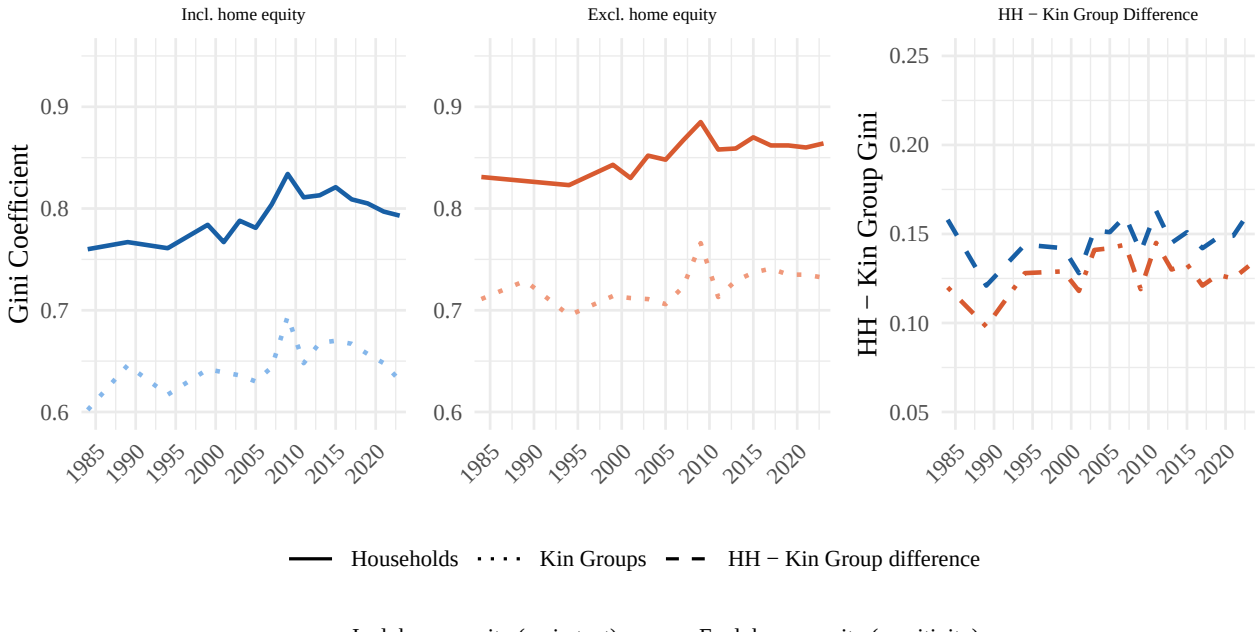
Appendix C illustrates extreme cases from simulated household and kin group data. The simulation demonstrates that kin group-level Gini coefficients are not necessarily lower than household-level Gini coefficients. The direction of the difference depends on the relationship between kin group size, the overall distribution of resources, and the selection of households into kin groups.

Appendix C1 shows the minimum possible difference between kin group and household Gini coefficients in the simulated data, while Appendix C2 shows the maximum possible difference. Curves show that Gini coefficients at the kin group level can exceed those of households. Appendix C3 illustrates that, while it is more likely for Gini coefficients at the household level to exceed Gini coefficients at the kin group level, whether this effect is observed depends on the combination between kin group size and income or wealth distribution.

Appendix D: Sensitivity Analyses — Gini Coefficients

Across a range of specifications, estimated inequality using the Gini is lower when kin groups are used as the unit of analysis instead of households. This pattern is not driven by race, nor does it disappear when expanding the sample, excluding home equity in measures of wealth, including negative wealth values, standardizing by household size, or applying weights. For each specification, results are recalculated and displayed alongside the main results.

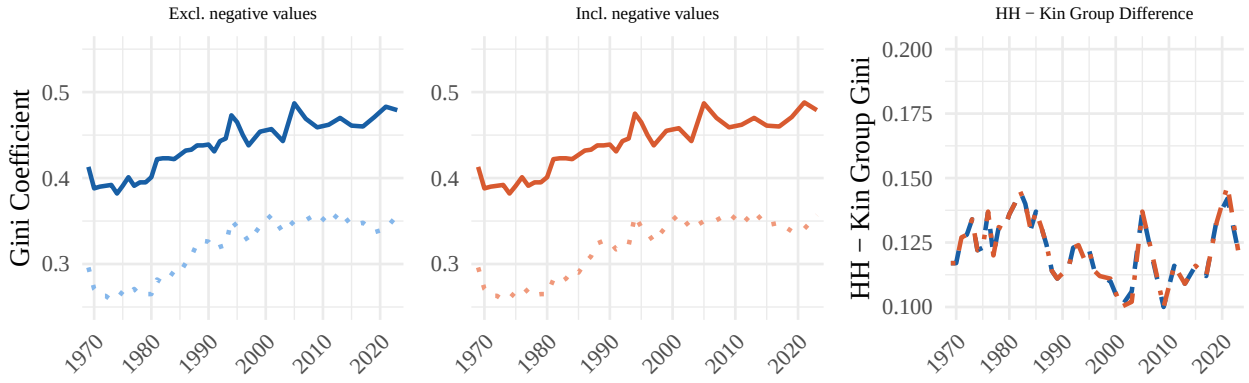
Appendix D1: Sensitivity — Wealth With and Without Home Equity



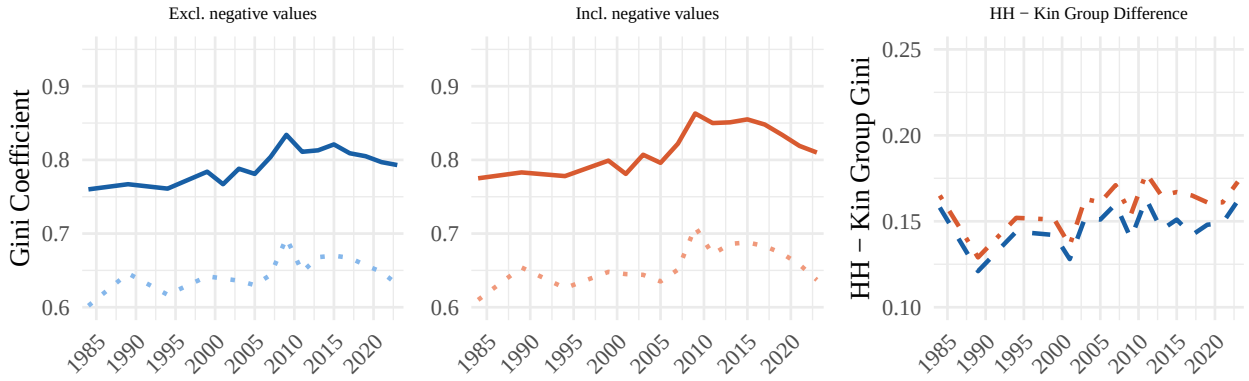
Results presented in the main text include home equity for measures of wealth. Appendix D1 shows Gini coefficients when home equity is excluded (middle graphs), and reports the difference between Gini coefficients at the household and kin group levels (graphs on the right). Excluding home equity makes estimated inequality lower and minimizes the observed difference between household and kin group Gini coefficients.

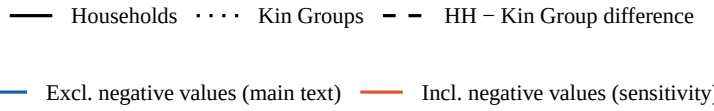
Appendix D2: Sensitivity — Negative Values

Panel A: Income



Panel B: Wealth

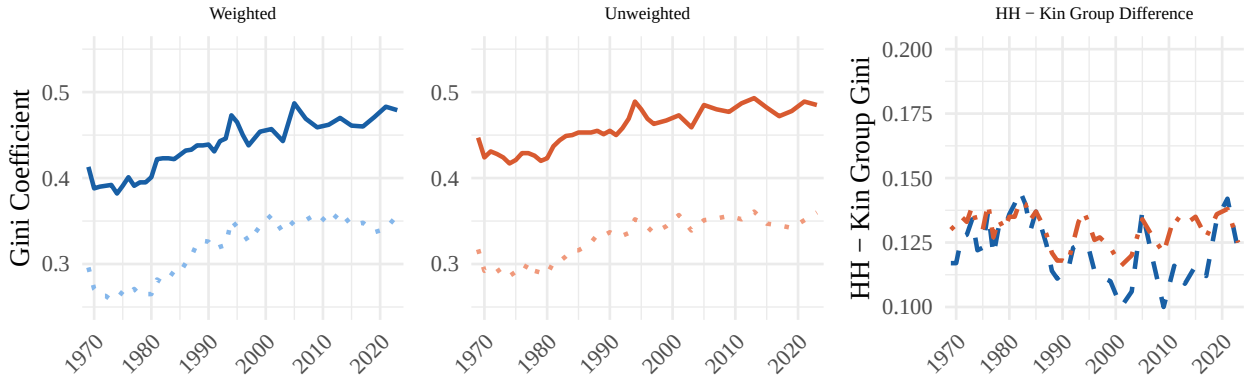




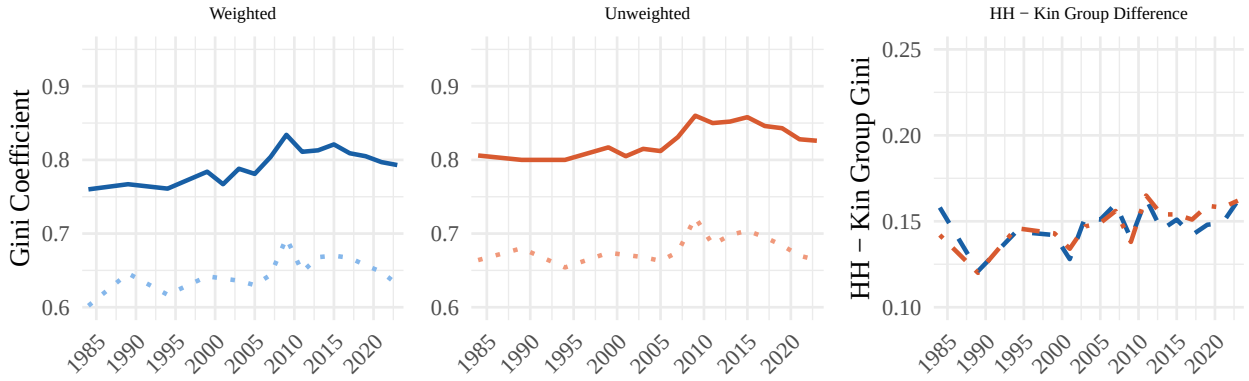
Results presented in the main text exclude negative values for income and wealth. Appendix D2 shows Gini coefficients when negative values are included (middle graphs), and reports the difference between Gini coefficients at the household and kin group levels (graphs on the right). Including negative values does not change estimated Gini coefficients at the household or kin group levels.

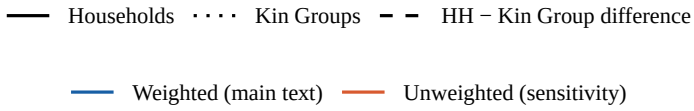
Appendix D3: Sensitivity — Weighting

Panel A: Income



Panel B: Wealth

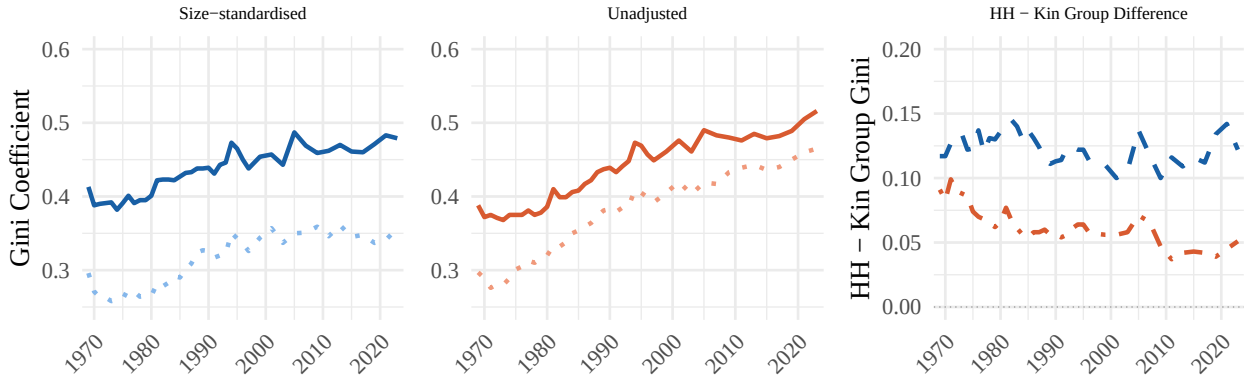




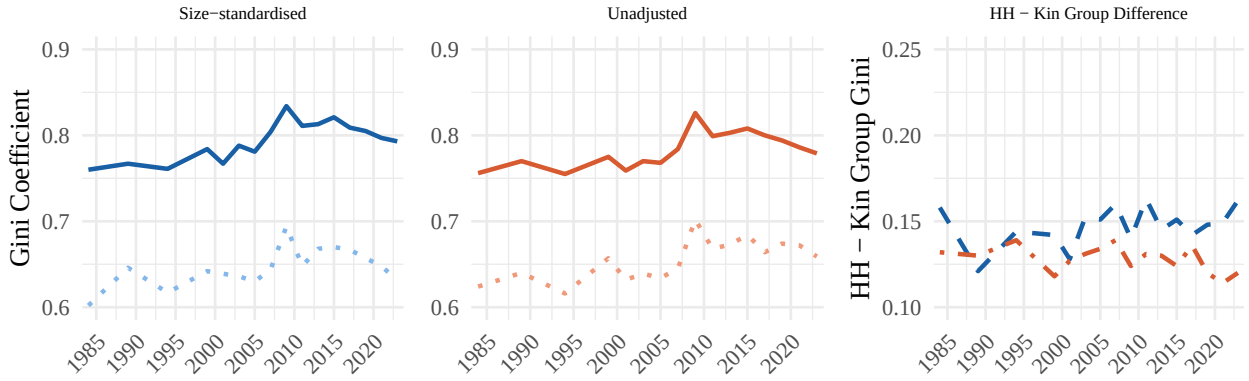
All results presented in the main paper are weighted. Household-level Gini coefficients are weighted using PSID-provided family weights, while kin group-level Gini coefficients are weighted using a constructed kin group weight based on the PSID family weight. Weighting does not substantially change the results.

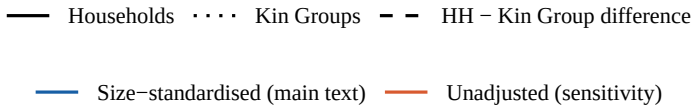
Appendix D4: Sensitivity — Size Standardization

Panel A: Income



Panel B: Wealth

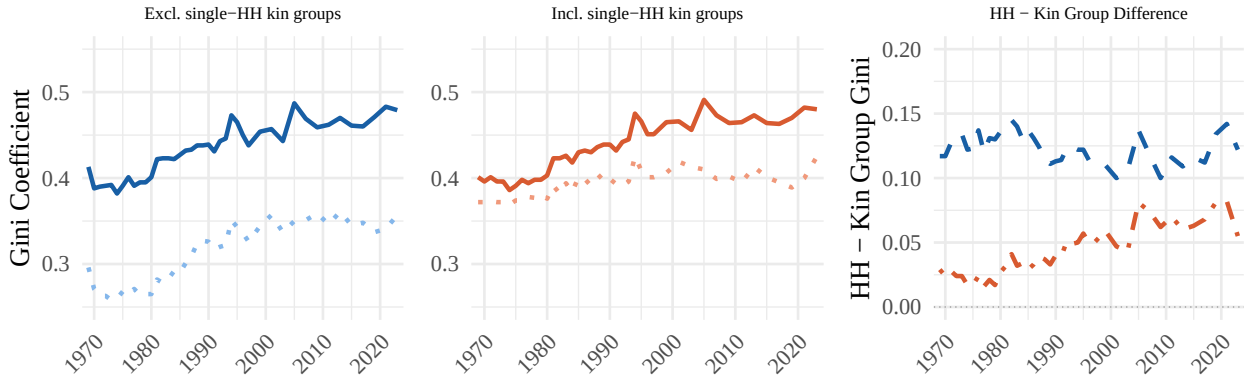




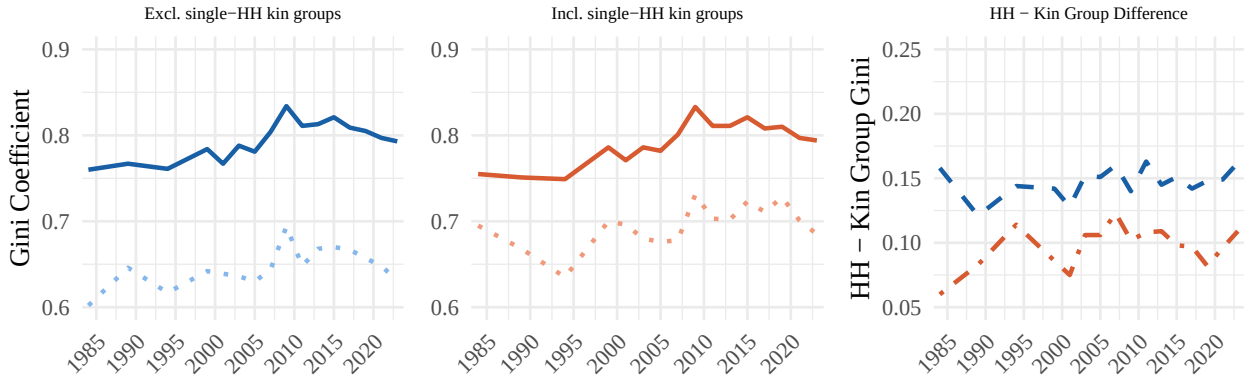
Results in the main text standardize income, wealth, and resulting Gini coefficients by household and kin group size. Estimated inequality is higher when results are not standardized by size and the gap between household and kin group Gini coefficients is smaller. However, Gini coefficients are higher at the kin group versus household level irrespective of whether measurements are adjusted by size.

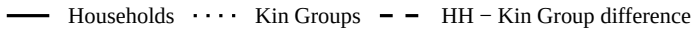
Appendix D5: Sensitivity — Single-Household Kin Groups

Panel A: Income



Panel B: Wealth





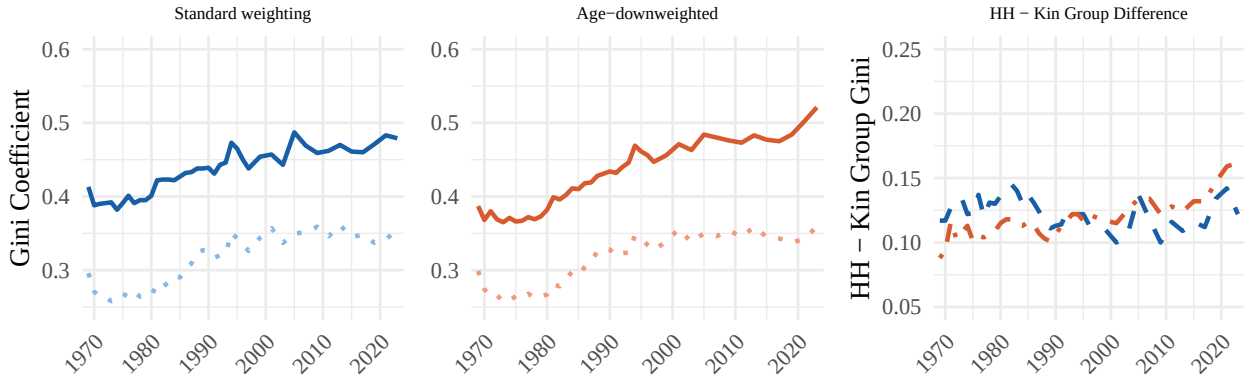


In reality, all households belong to kin groups that consist of multiple households. However, some households in the PSID do not have kin groups due to the survey design and attrition. The survey is genealogical, meaning that initial households only have measurable kin groups once their descendants respond to the survey. Attrition also might limit responses from some households, thereby erasing kin groups for the households they are connected to.

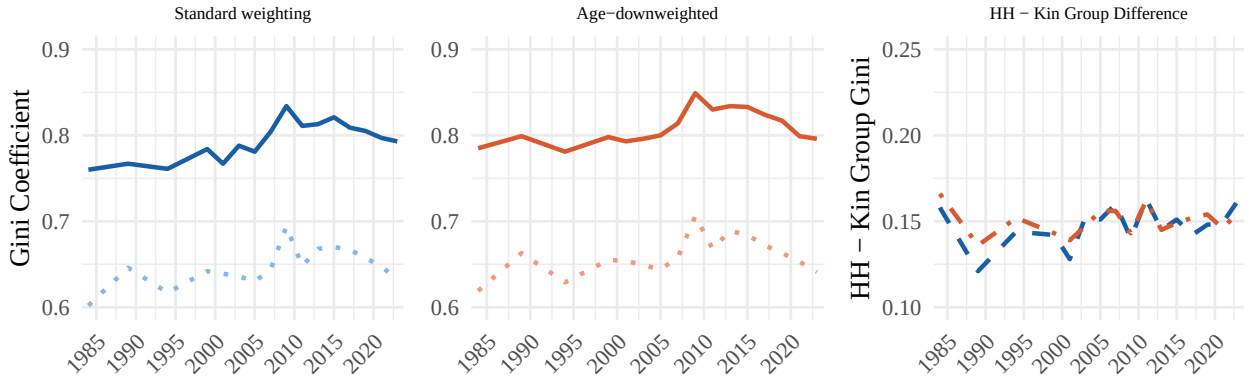
Because these limitations are imposed by the survey design, not social reality, we exclude household-years where households are alone in their kin group from both household and kin group measurements. The graphs on the left side show this. When including these single-household kin groups, or "kin groupless" households, the difference in Gini coefficients at the household and kin groups levels is minimized but not eliminated. The graphs in the middle show Gini coefficients when single-household kin groups are included. The graphs on the right side show the difference between household and kin group (HH - kin group) Gini coefficients for each group. Differences in estimated inequality at the kin group and household levels are smaller when including kin groupless households, however, they still exist and are increasing over time.

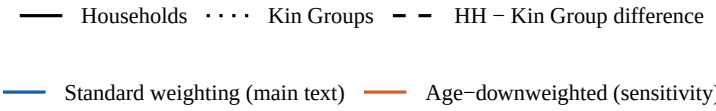
Appendix D6: Sensitivity — Age Weighting

Panel A: Income



Panel B: Wealth

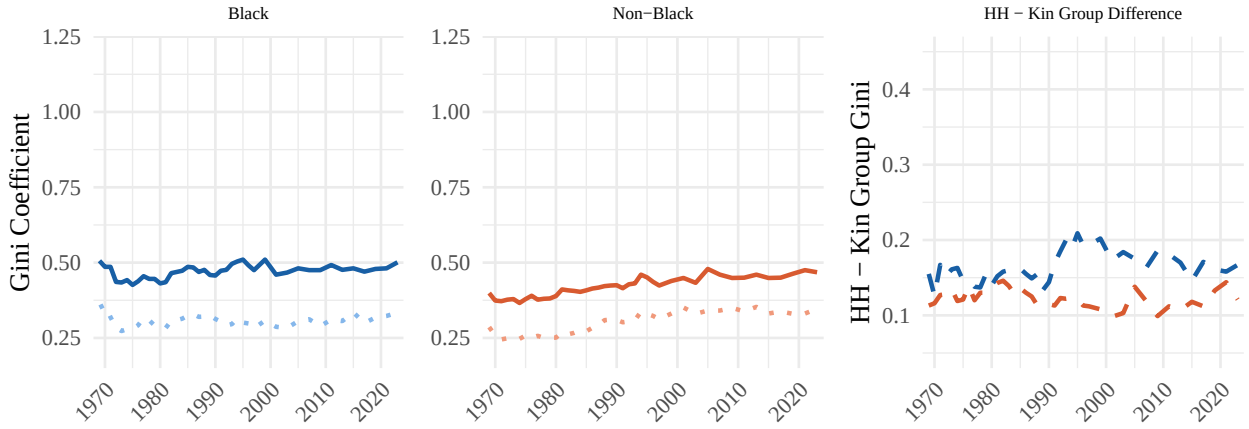




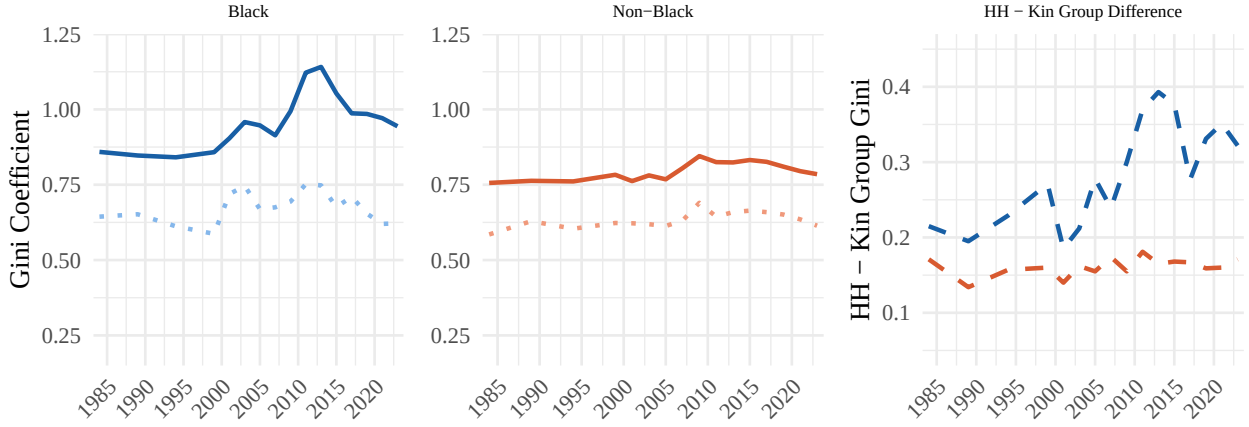
Results in the main text use standard PSID family and kin group weights. Appendix D6 shows Gini coefficients when these weights are downweighted for older households and kin groups (middle graphs): each weight is multiplied by the sample mean age divided by the unit's mean age. Gini coefficients remain higher at the household level than at the kin group level under age weighting.

Appendix D7: Gini Coefficients by Race Subgroup

Panel A: Income



Panel B: Wealth (incl. home equity)



Appendix D7 shows that the differences between Gini coefficients at the household and kin group level are not driven by racial subgroups. Graphs on the left side show Gini coefficients for Black households and kin groups. Graphs in the middle show Gini coefficients for non-Black households and kin groups. The graphs on the right side show the difference between household and kin group (HH - kin group) Gini coefficients for each group. Gini coefficients for income (Panel A) and wealth (Panel B) are higher at the kin group level for both Black and non-Black kin groups, indicating that inequality is lower when estimated at the kin group level across racial groups.

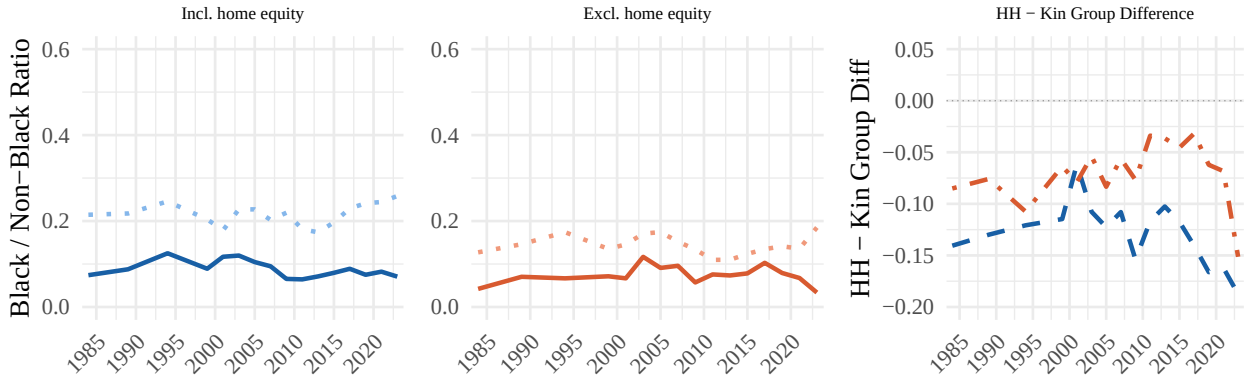
Appendix E: Sensitivity Analyses — Race Ratios

Across a range of specifications, estimated median racial wealth gaps are lower when kin groups are used as the unit of analysis instead of households. This pattern does not disappear when expanding the sample, excluding home equity in measures of wealth, including negative wealth values, standardizing by household size, or applying weights. For each specification, results are recalculated and displayed alongside the main results.

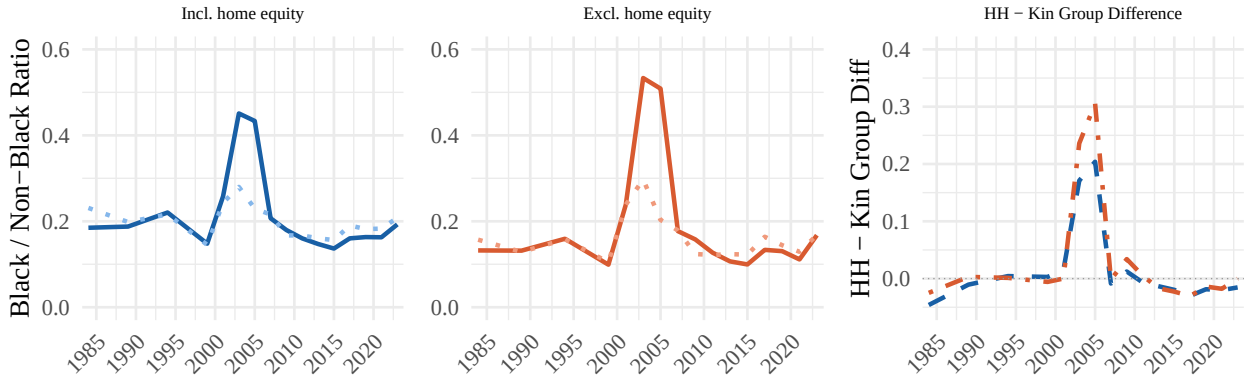
Mean racial wealth gaps are about the same at the household and kin group levels. This finding is also robust across each specification. Appendix E7 explains the difference between the mean and median wealth ratios and why median racial wealth gaps are presented.

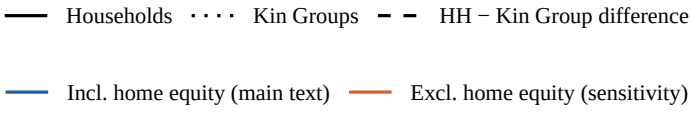
Appendix E1: Race Ratios — Wealth With and Without Home Equity

Panel A: Median wealth ratio



Panel B: Mean wealth ratio

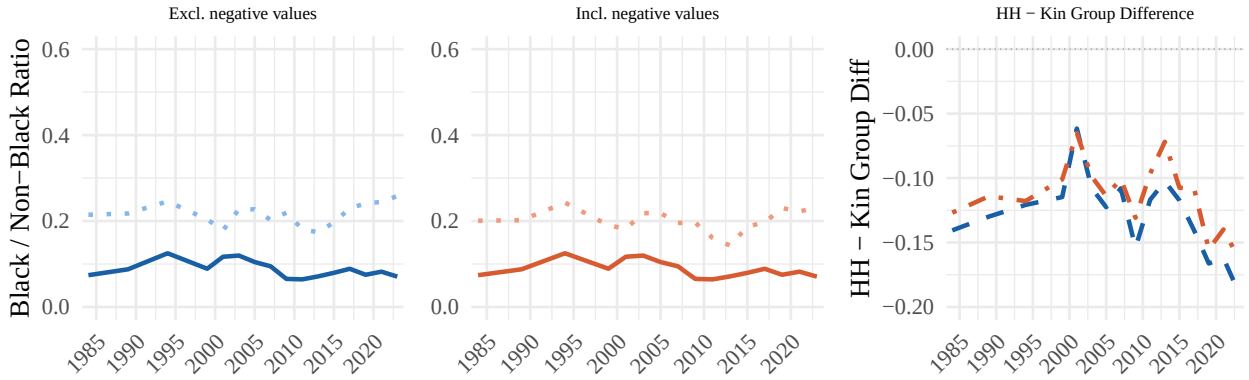




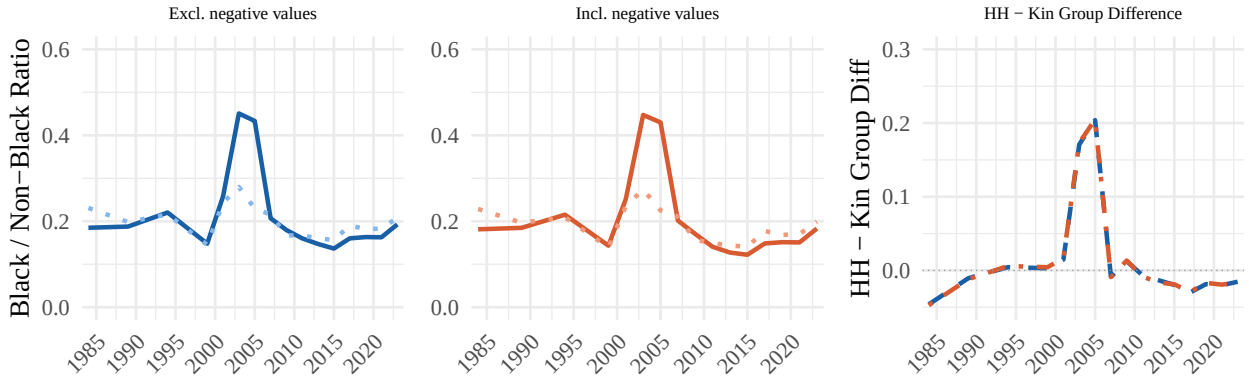
Appendix E1 compares racial wealth ratios when home equity is included (main specification, left panels) versus excluded (middle panels). Panel A shows median wealth ratios and Panel B shows mean wealth ratios. The median racial wealth gap between kin groups, versus households, shows less inequality irrespective of whether wealth measures include home equity.

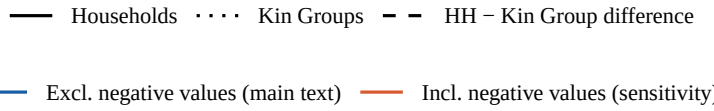
Appendix E2: Race Ratios — Negative Values

Panel A: Median wealth ratio



Panel B: Mean wealth ratio

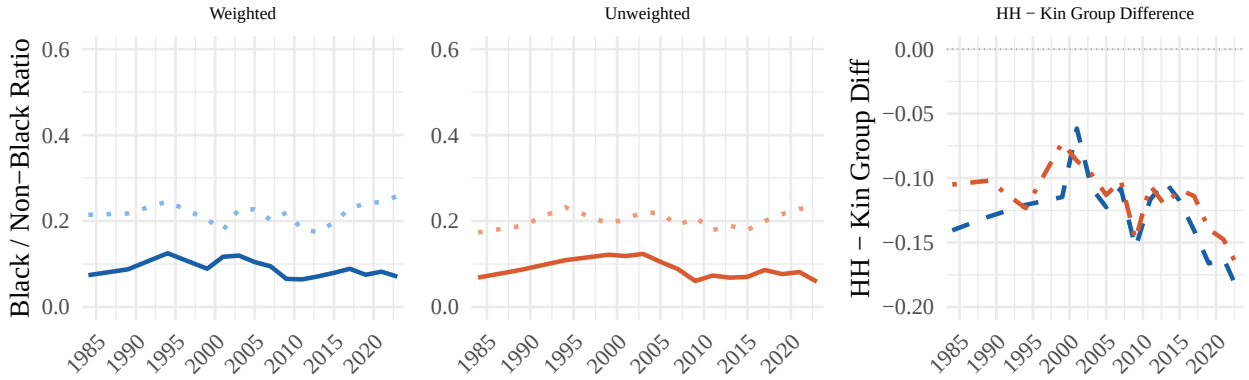




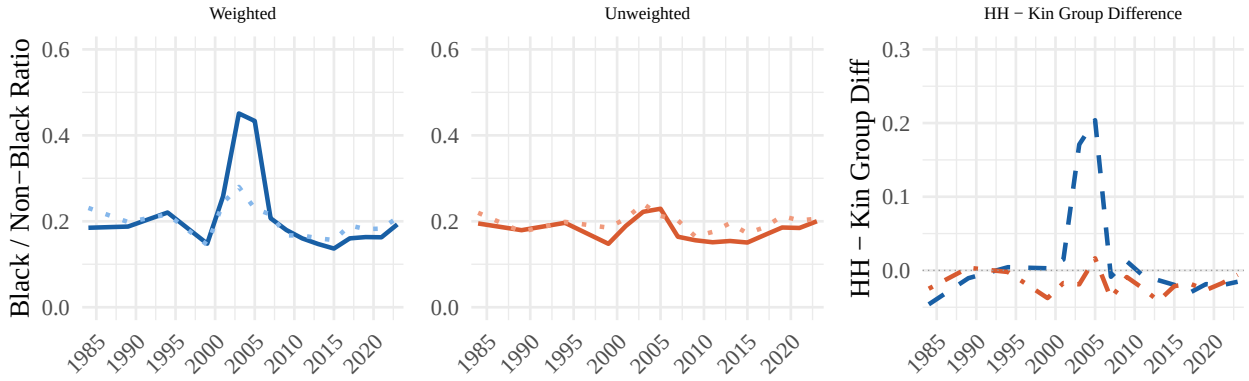
Appendix E2 shows Black / Non-Black wealth ratios when negative wealth values are excluded (main specification, left panels) versus included (middle panels). Panel A shows median wealth ratios and Panel B shows mean wealth ratios. Including negative values does not change the estimated race ratios.

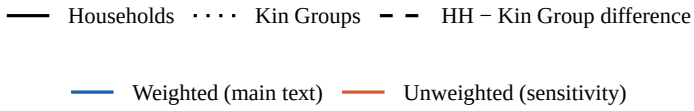
Appendix E3: Race Ratios — Weighting

Panel A: Median wealth ratio



Panel B: Mean wealth ratio

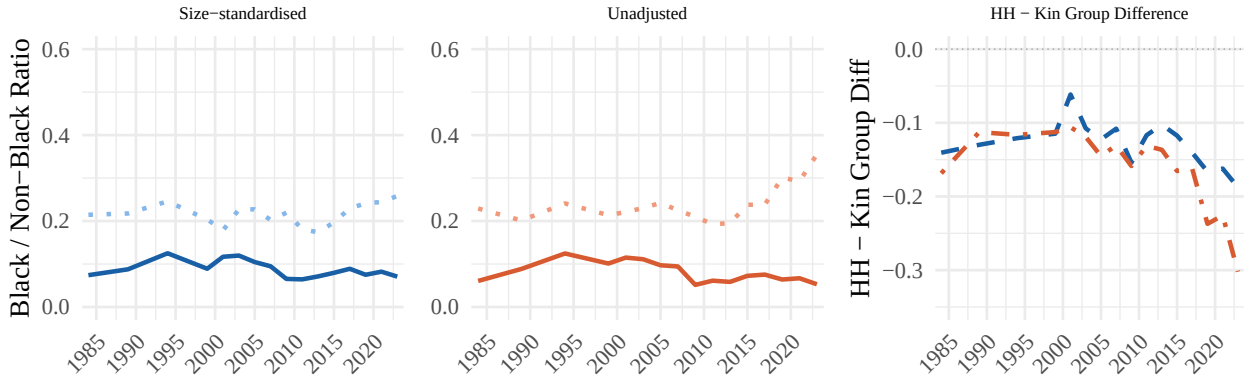




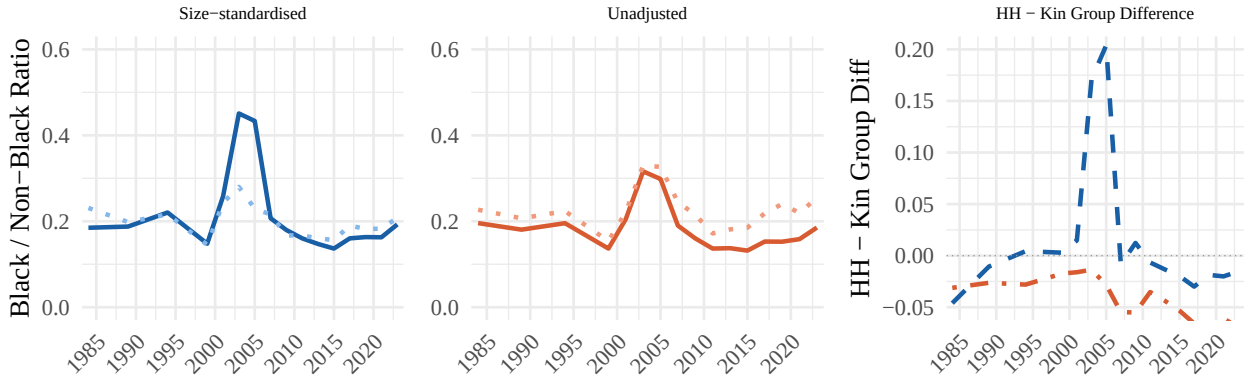
Appendix E3 shows Black / Non-Black wealth ratios using weighted estimates (main specification, left panels) versus unweighted estimates (middle panels). Panel A shows median wealth ratios and Panel B shows mean wealth ratios. Weighting does not substantially change the estimated race ratios.

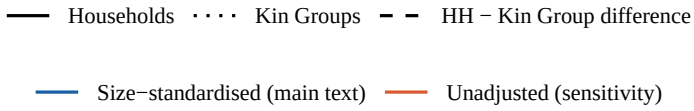
Appendix E4: Race Ratios — Size Standardization

Panel A: Median wealth ratio



Panel B: Mean wealth ratio

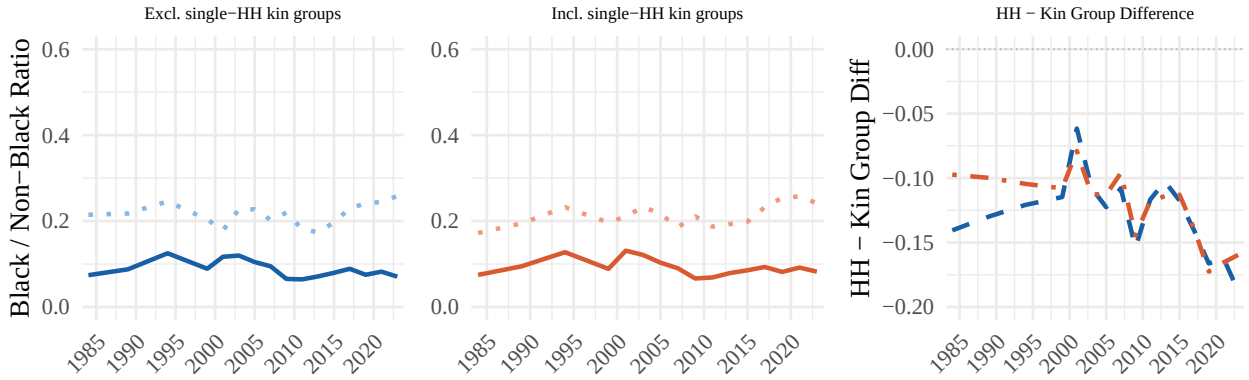




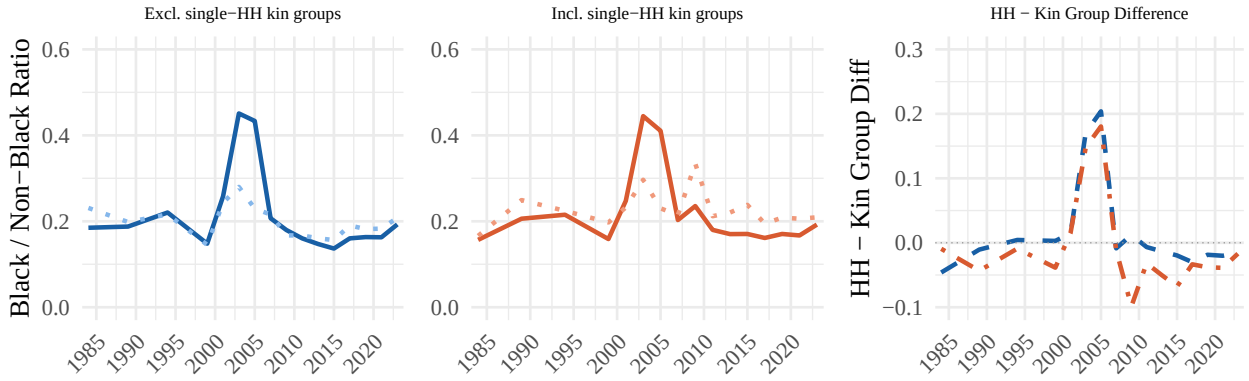
Appendix E4 compares Black / Non-Black wealth ratios when wealth is size-standardized by household and kin group size (main specification, left panels) versus unadjusted (middle panels). Panel A shows median wealth ratios and Panel B shows mean wealth ratios. Standardizing by size does not substantially change the estimated median wealth race ratios from 1984–2015. From 2015–2023, the difference between median wealth gaps at the kin group versus household level widen.

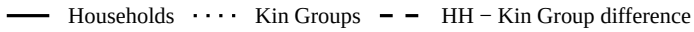
Appendix E5: Race Ratios — Single-Household Kin Groups

Panel A: Median wealth ratio



Panel B: Mean wealth ratio

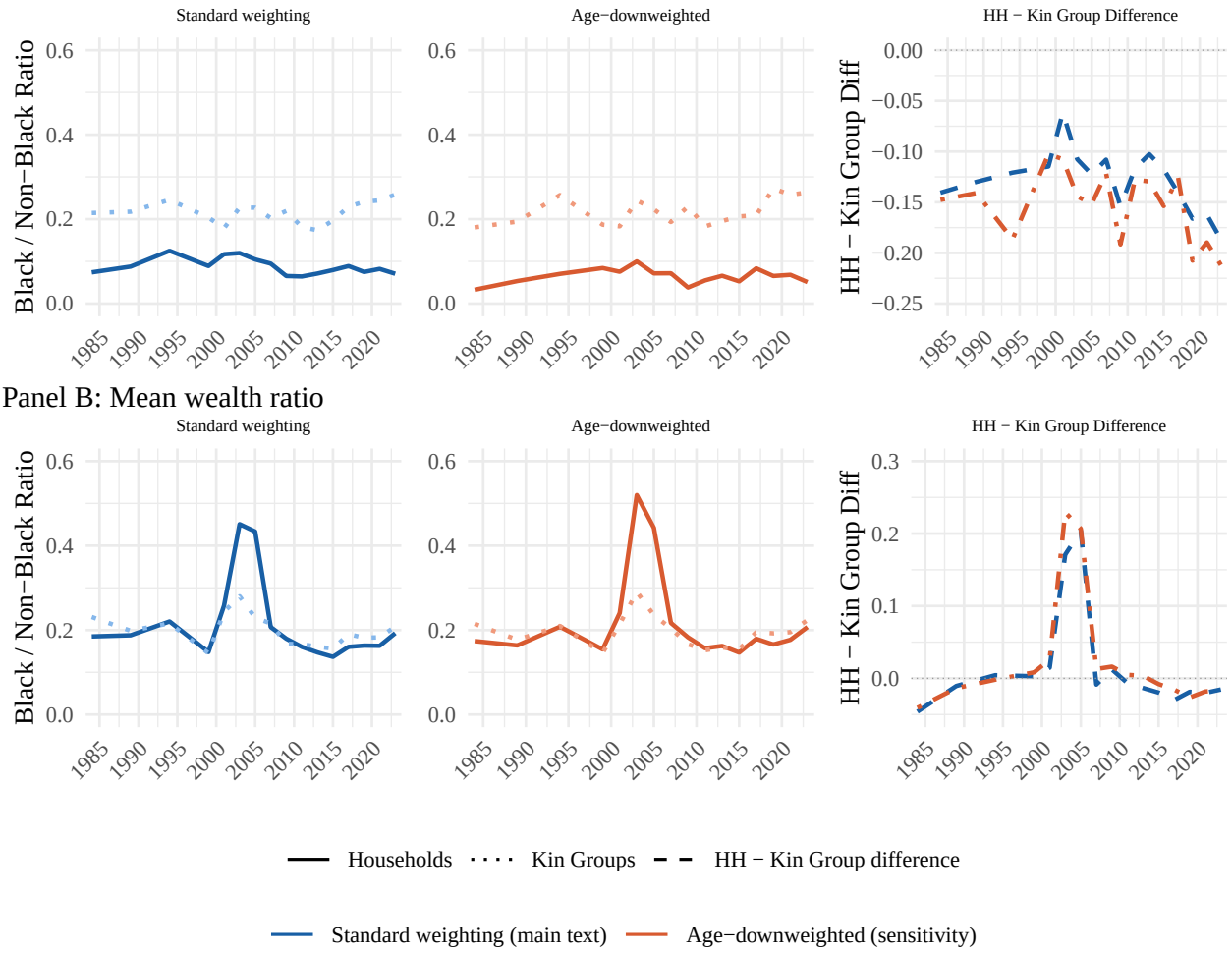






Appendix E5 shows Black / Non-Black wealth ratios when single-household kin groups are excluded (main specification, left panels) versus included (middle panels). Panel A shows median wealth ratios and Panel B shows mean wealth ratios. Including single-household kin groups does little to change racial wealth gaps using median or mean wealth (right panels). In particular, the median racial wealth gap between kin groups, versus households, shows less inequality irrespective of whether single-household kin groups are included.

Appendix E6: Race Ratios — Age Weighting

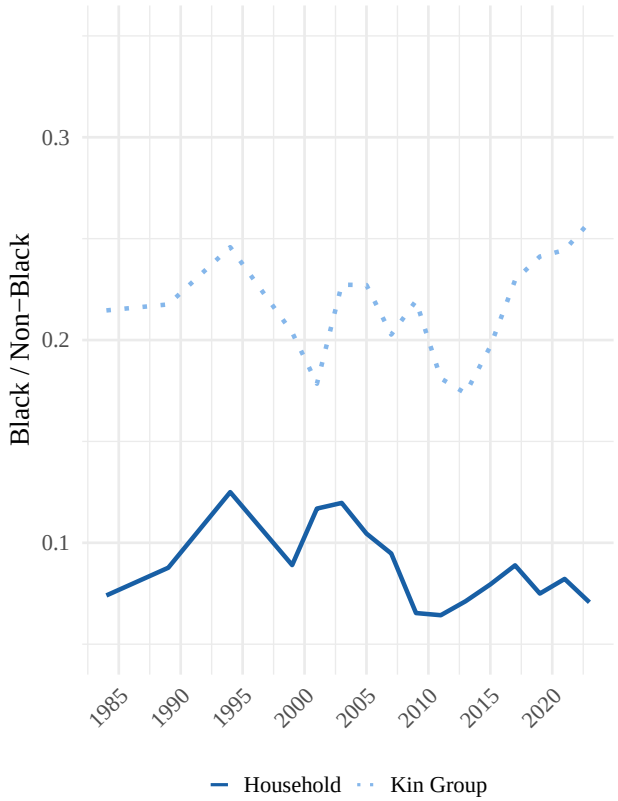


Appendix E6 shows Black / Non-Black wealth ratios using standard weights (main specification, left panels) versus weights that downweight older households and kin groups (middle panels). Each weight is multiplied by the sample mean age divided by the unit's mean age. Panel A shows median wealth ratios and Panel B shows mean wealth ratios.

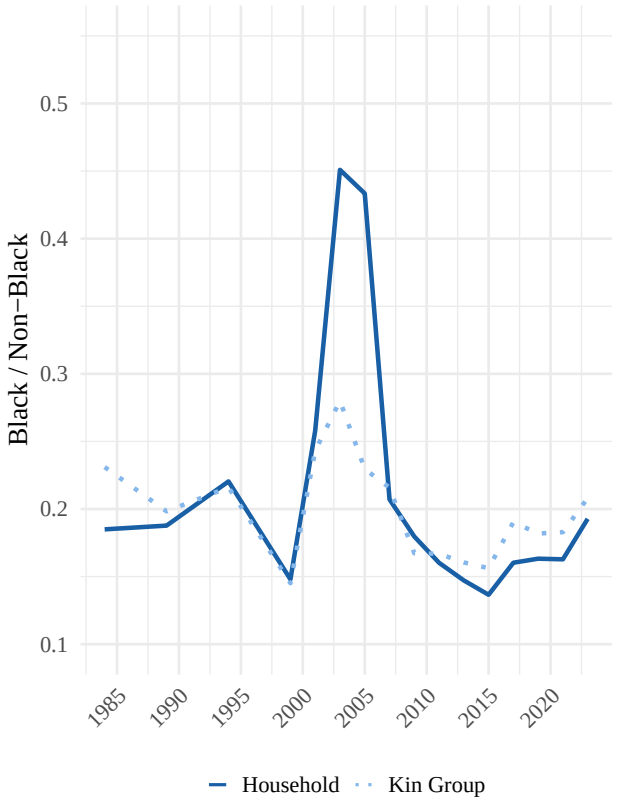
Appendix E7: Mean and Median Wealth Ratios

Appendix E7. Black / Non−Black Wealth Ratios: Households vs. Kin Groups

Panel A: Median wealth ratio



Panel B: Mean wealth ratio



Appendix E7 reports Black / non-Black wealth ratios using weighted means for both households and kin groups. Panel A reproduces the median wealth ratios (as in Figure 3 of the main paper), while Panel B presents mean wealth ratios. Using medians, the racial wealth gap differs between households and kin groups (Panel A), whereas using means, the two are about the same (Panel B).

Race ratios using mean wealth, when pooled across all years, show that Black households hold 19 cents for every dollar of wealth held by non-Black households, while Black kin groups hold 18 cents for every dollar held by non-Black kin groups. The convergence in mean-based estimates is driven by a small number of high-wealth Black households in the PSID, particularly in 2004–2005, which skew household-level averages. These results highlight the sensitivity of mean-based measures to outliers, which is why we present race ratios based on median wealth in the main text.

Appendix F: Household and Kin Group Size and Age

Appendix F1: Characteristics of Households and Kin Groups, 1984 and 2023

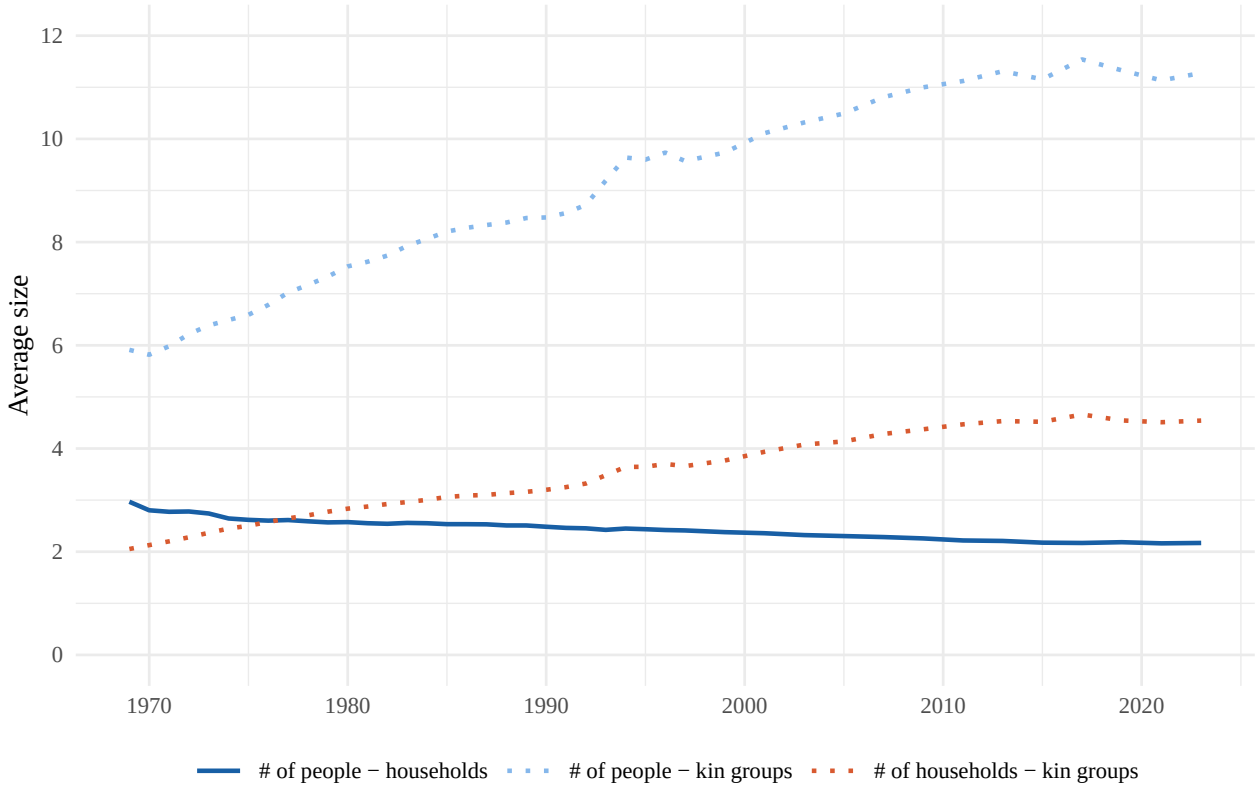
Appendix F1. Characteristics of Households and Kin Groups, 1984 and 2023

	1984		2023	
	Households	Kin Groups	Households	Kin Groups
Average age (years)	29.7	28.9	39.6	37.3
Average # of people	2.55	8.06	2.17	11.28
Average # of households	—	3.01	—	4.54
Median income	$67,610	$215,901	$68,280	$368,636
Median wealth	$67,451	$337,958	$131,000	$986,500
N (unit-years)	5,496	1,731	8,490	1,610
N Black (unit-years)	2,164	643	3,550	513

Note: Average age, number of people, and number of households are based on the income sample (1969-2023), which contains 2,906 unique kin groups. Median wealth is based on the wealth sample (1984-2023) and includes home equity. Size and median values are weighted using PSID family and constructed kin group weights (see Methods). Pooled across all years, the wealth sample contains 122,069 household-years and 26,655 kin group-years (46,287 and 8,470 Black, respectively) from 2,636 unique kin groups, with an average size of 2.27 individuals per household and 10.56 individuals (4.18 households) per kin group, and an average age of 36.3 for households and 34.1 for kin groups. Estimates are adjusted for inflation and expressed in 2023 dollars.

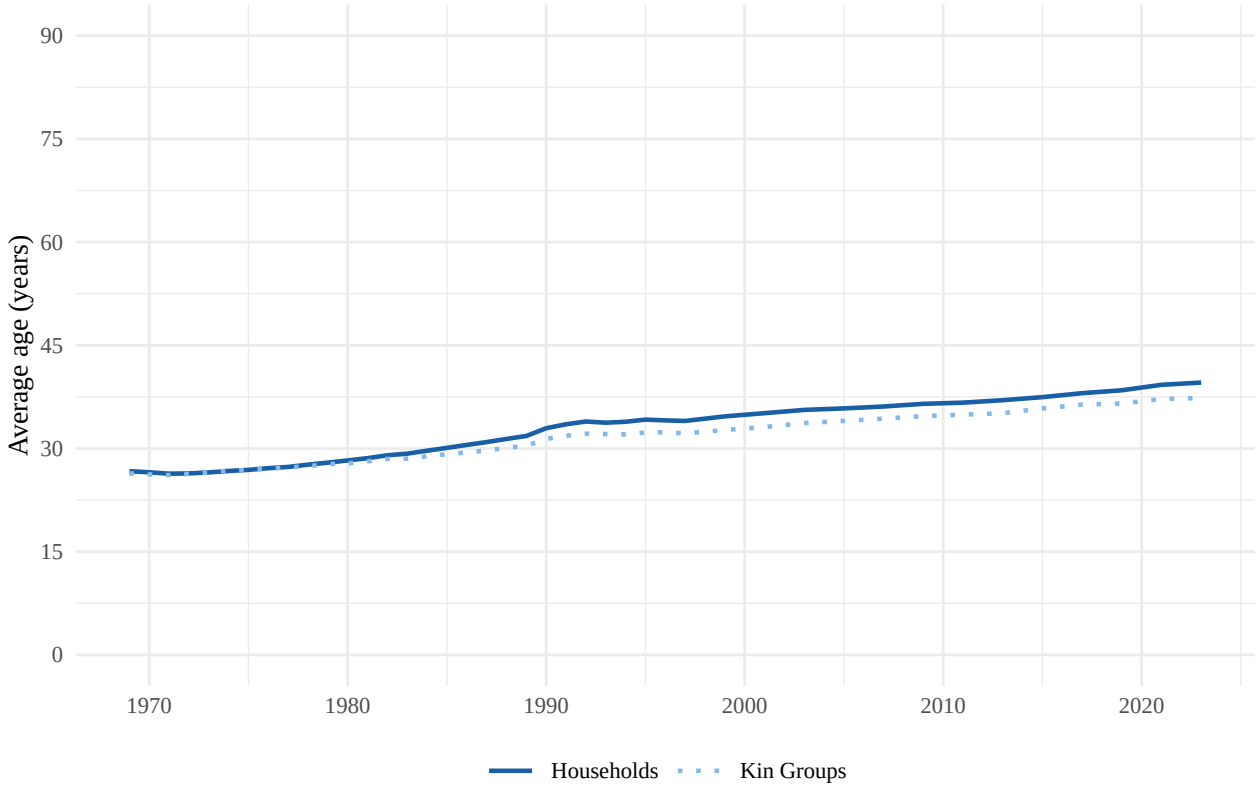
Appendix F1 reports the household and kin group characteristics shown in Table 1 of the main paper separately for 1984 and 2023.

Appendix F2: Household and Kin Group Size Over Time



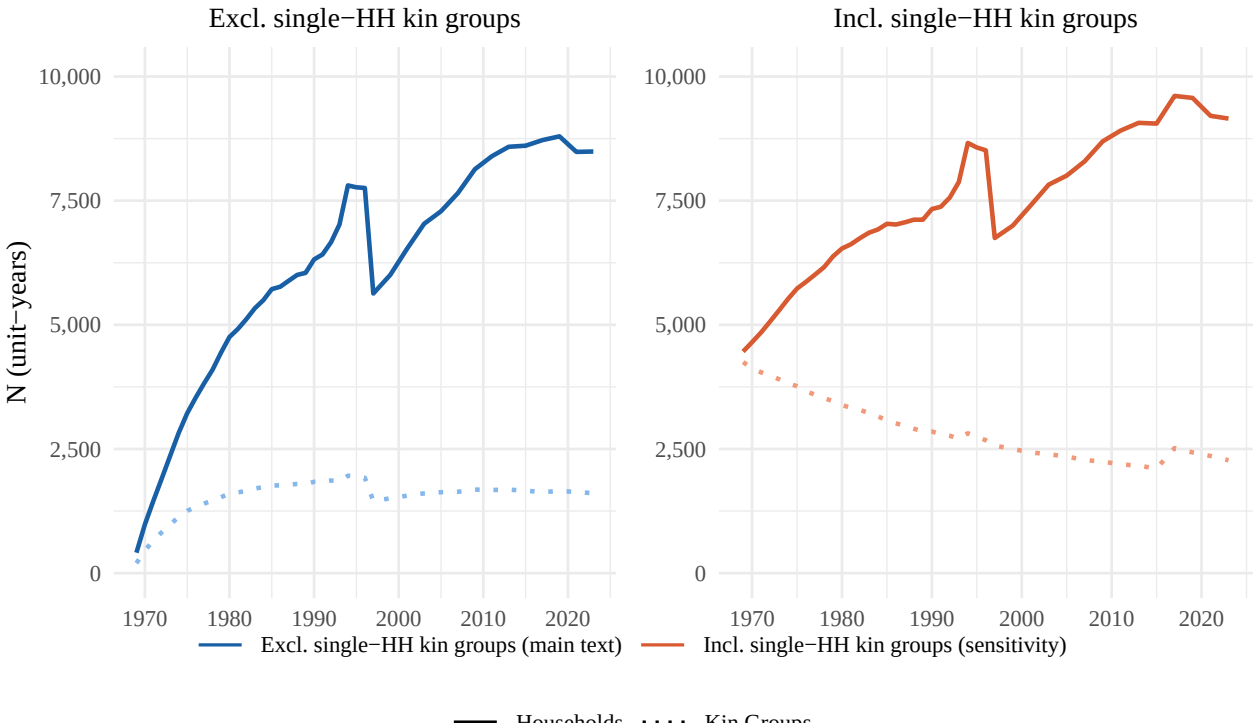
Appendix F2 plots the weighted average number of people per household and per kin group (blue), and the weighted average number of households per kin group (orange), in each year of the income sample.

Appendix F3: Average Age of Households and Kin Groups Over Time



Appendix F3 plots the average age of households and kin groups in each year of the income sample.

Appendix F4: Sample Size With and Without Single-Household Kin Groups



Appendix F4 plots the number of households and kin groups in each year when single-household kin groups are excluded (main specification, left panel) versus included (right panel).

[1] FRED. GINI Index for the United States. Federal Reserve Bank of St. Louis.; 2025. https://fred.stlouisfed.org/series/SIPOVGINIUSA

[2] Census. Table F-4. Gini Indexes of Families by Race and Hispanic Origin of Householder. U.S. Census Bureau; 2025. https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-income-inequality.html

[3] Wolff E. The Size Distribution of Wealth in the United States: A Comparison among Recent Household Surveys. In: Wealth, Work, and Health: Innovations in Measurement in the Social Sciences. Panel Study of Income Dynamics; 209 AD.

[4] Pfeffer FT, Danziger S, Schoeni RF. Wealth Disparities before and after the Great Recession. The Annals of the American Academy of Political and Social Science. 2013;650(1):98–123. doi:10.1177/0002716213497452

[5] Aaberge R. Gini's nuclear family. Journal of Economic Inequality. 2007 [accessed 2025 Dec 16];5(3):305–322. https://research.ebsco.com/linkprocessor/plink?id=183a0f60-6b65-3238-9faf-59d099fff528. doi:10.1007/s10888-006-9050-8